

Toby Jennings

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PERSONAL PROFILE

Final-year Computer Science student on track for a First, seeking a Software Engineering graduate role. Proven engineer through fintech and data engineering internships: building a secure file-transfer platform at NatWest Group and reducing a 40-step ETL process to 11 at Technicus. Two hackathon wins and strong foundations in algorithms and programming.

EDUCATION

BSc Computer Science, University of Southampton

2024 - 2027

Year 1 average: 1st, Year 2 average: 1st, on track for a 1st in final year.

Excelled in modules such as Operating Systems, System Architecture, Algorithmics, and Networks & Security.

A-Levels, George Abbot School

2022 - 2024

Maths (A*), Further Maths (A), Computer Science (A).

EXPERIENCE

Software Engineer Intern - NatWest Group, London

Jun 2026 - Aug 2026

- Designed and built a secure file-transfer solution for the bank's internal network, overcoming a long-standing issue of file size restrictions by increasing the maximum transferable file size by ~400x.
- Built an automated request and approval pipeline (Java Spring Boot, React), reducing wait time from days to minutes.
- Added automated malware scanning and audit logging for every transfer, replacing the previous manual process.

Junior Data Engineer - Technicus Ltd, London

Jun 2025 - Sep 2025

- Built a config-driven Python tool that automated dbt transformations through the full pipeline and triggered PowerShell scripts using the Salesforce Bulk Loader API, reducing a 40-step manual ETL process to 11 steps.
- Worked with senior engineers on a data migration dry run into pre-Production, resolving issues ahead of go-live.

Customer Service Assistant (part-time) - Guildford Spectrum

Aug 2022 - Sep 2025

- Worked part-time at the front desk in a busy leisure centre, managing customer bookings, payments, and enquiries.
- Efficiently handled difficult customer complaints, often turning them around and creating positive outcomes.

SKILLS

- Programming Languages: Python, C++, C, Java, Bash
- Embedded Systems: ESP32 (C++), PlatformIO, Arduino
- Tools & Platforms: Daily use of Linux, Git, Jupyter, CMake, FastAPI, Spring Boot, SQL, MongoDB, AWS

PROJECTS

1st Place, ElevenLabs Track at SotonHack 2026 (github.com/kiwifruit0/Echo)

- Created a voice-based social app, combining Reddit-style forums with BeReal-style daily voice notes between friends.
- Built a low-latency voice pipeline in Python FastAPI using ElevenLabs for TTS/STT, storing audio in MongoDB Atlas.
- Designed the frontend in React and Tailwind, featuring reactive 3D visuals and a 3D connected friend graph.
- Integrated Gemini for intent detection, summarisation, and interest classification to personalise each user's feed.

1st Place, Soton Data Science x WECS Hackathon (github.com/Ryan-Shino/WECSHackathonProject)

- Built a location-aware web app to gamify real-world activity and incentivise socialising, learning, and exercise.
- Developed a Python FastAPI backend with a React + Tailwind frontend integrated via REST APIs.
- Implemented real-time geolocation processing with Mapbox API integration for classifying users' location.
- Designed a progression system with category-based stat tracking and competitive leaderboards.

Machine Learning Grade Predictor (github.com/kiwifruit0/grade_predictor)

- Built a prototype application to predict students' grades from academic and non-academic attributes, helping educators identify students who may need additional support.
- Engineered features and built a custom regression model in Python, trained on an open-source dataset.
- Tuned the regression algorithm and evaluated model performance, improving ~15% over a baseline linear model.
- Surfaced the most influential attributes behind student outcomes, giving educators visibility into key predictors.